

Configuration & Troubleshooting

1. Basic Router Configuration

Access Router CLI

```
Router> enable
Router# configure terminal
Router(config)#
```

Set Hostname

```
Router(config)# hostname R1
```

Result:

```
R1(config)#
```

Configure Interface IP Address

Example:

```
R1(config)# interface gigabitEthernet0/0
R1(config-if)# ip address 192.168.1.1 255.255.255.0
R1(config-if)# no shutdown
```

Explanation:

```
interface → select port
ip address → assign IP
no shutdown → enable interface
```

Check Interface Status

```
show ip interface brief
```

Example output:

Interface	IP Address	Status
Gig0/0	192.168.1.1	up
Gig0/1	unassigned	down

2. Basic Switch Configuration

Enter configuration mode:

```
Switch> enable  
Switch# configure terminal
```

Set Hostname

```
Switch(config)# hostname S1
```

Assign IP to Switch (Management)

```
S1(config)# interface vlan 1  
S1(config-if)# ip address 192.168.1.2 255.255.255.0  
S1(config-if)# no shutdown
```

Default gateway:

```
S1(config)# ip default-gateway 192.168.1.1
```

3. Configure Static Route

Static route tells router **how to reach remote network**.

Example:

```
R1(config)# ip route 10.0.0.0 255.255.255.0 192.168.1.2
```

Meaning:

```
Destination network → 10.0.0.0
```

```
Next hop → 192.168.1.2
```

4. Configure Default Route

Used for **unknown networks (internet)**.

```
R1(config)# ip route 0.0.0.0 0.0.0.0 192.168.1.1
```

5. Verify Routing Table

```
show ip route
```

Example:

```
C 192.168.1.0/24 directly connected
```

```
S 10.0.0.0/24 via 192.168.1.2
```

```
S* 0.0.0.0/0 via 192.168.1.1
```

6. Troubleshooting Commands

These commands are **very important in CCNA and interviews**.

Check Interface Status

```
show ip interface brief
```

Check Routing Table

```
show ip route
```

Check MAC Address Table

```
show mac address-table
```

Check ARP Table

```
show arp
```

Test Connectivity

Ping

```
ping 192.168.1.10
```

Used to check if a device is reachable.

Traceroute

```
traceroute 8.8.8.8
```

Shows **path** packets take to destination.

7. Common Network Problems

1. Interface Shutdown

Problem:

```
Interface down
```

Fix:

```
no shutdown
```

2. Wrong IP Address

Example:

```
PC → 192.168.1.10  
Router → 192.168.2.1
```

Different networks → communication fails.

3. Wrong Subnet Mask

Incorrect mask may cause routing problems.

Example:

```
255.255.255.0 vs 255.255.0.0
```

4. Missing Default Gateway

Without gateway, device **cannot reach other networks**.

5. VLAN Misconfiguration

Devices in different VLANs cannot communicate without routing.

8. Basic Troubleshooting Method

Network engineers usually follow this process:

1. Identify the problem
 2. Gather information
 3. Check physical connections
 4. Check IP configuration
 5. Test connectivity (ping)
 6. Check routing table
 7. Fix configuration
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9. Important Interview Questions

Q1. How do you check interface status?

Answer:

```
show ip interface brief
```

Q2. How do you check routing table?

```
show ip route
```

Q3. How do you test connectivity between devices?

```
ping
```

Q4. What command shows MAC address table?

```
show mac address-table
```

Q5. What command enables a router interface?

```
no shutdown
```

Q6. What command shows ARP entries?

```
show arp
```

Q7. What is the difference between ping and traceroute?

Ping → tests connectivity

Traceroute → shows the path packets take

Quick Revision Commands

```
show ip interface brief  
show ip route  
show arp  
show mac address-table  
ping  
traceroute
```

One Line Summary

Configuration → setting up router/switch

Troubleshooting → finding and fixing network problems